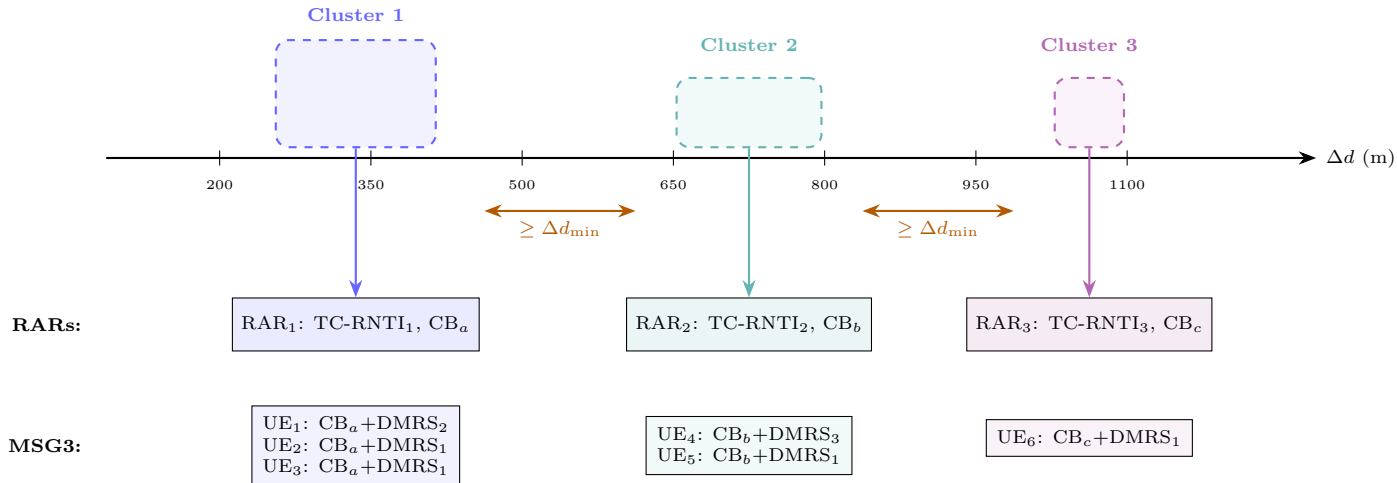


ToA Clustering by Differential Distance

$$\Delta d = d_{\text{slant}} - h_{\text{sat}}$$



Intra-cluster behavior:

- All UEs inherit the same SCMA layer
- Separation relies on pilot-sequence (DMRS) diversity
- Two UEs picking the same pilot cause a *pilot collision* (both fail)
- UEs sharing a layer are *contenders*

Inter-cluster behavior:

- Each cluster gets a distinct SCMA layer and uplink resource
- A UE alone in its SCMA layer is *scheduled*: the eNB knows its context and decodes the signal
- UE₆ illustrates the scheduled case